

Course Code	Course Name	Teaching Scheme (Contact Hours)			Credits Assigned			
		Theory	Pract.	Tut.	Theory	Pract.	Tut.	Total
2014111	Computational Theory	3	-	-	3	-	-	3

Course Code	Course Name	Theory					Term work	Pract / Oral	Total
		Internal Assessment			End Sem Exam	Exam Duration (in Hrs)			
		Test 1	Test 2	T1 +T2					
2014111	Computational Theory	20	20	40	60	2	--	--	100

Course Objectives:	1) To acquire conceptual knowledge of grammar and languages. 2) To understand the relation between Regular Language and Finite Automata. 3) To understand the language hierarchy, CFG and CFL. 4) To design a PDA equivalent to a given context-free grammar/language. 5) To learn the principles of computation by designing a Turing Machine 6) To infer the knowledge of undecidable and NP class problems.
Course Outcomes:	Upon completion of the course, the learners will be able to: 1) Use TCS theory to design regular expressions that represent regular languages. 2) Design, analyze, and optimize Finite Automata for language recognition. 3) Design Regular and Context Free Grammars and learn to simplify the CFG. 4) Design PDA for a given context-free grammar or language and enumerate its applications. 5) Design Turing machines as generators, deciders, and acceptors for various computational tasks. 6) Understand and utilize problem classification techniques for problem analysis.

Detailed Contents:

Sr. No.	Name of Module	Detailed Content	Hrs	CO Mapping
0	Prerequisite	Basic Mathematical Fundamentals: Sets, Logic, Relations, Functions, Discrete Structures.		
I	Basics Concepts and Regular Languages	Importance of TCS, Alphabets, Strings, Languages	1	CO1
		Regular operations, Regular Expression, Arden's theorem, RE Applications, Regular Language, Closure properties. Decision properties of RLs, Pumping lemma for RLs.	5	CO1
		Self-learning Topics: RE in text search and replace, Application of Regular Languages in Compiler Design, Text Processing, and Natural Language Processing (NLP).		
II	Finite Automata	Finite Automata (FA) & Finite State machine (FSM).	1	CO2
		Deterministic Finite Automata (DFA) and Nondeterministic Finite Automata (NFA): Definitions, transition diagrams and Language recognizers, Equivalence between NFA with and ϵ -transitions, NFA to DFA Conversion, Minimization of DFA, FSM with output: Moore and Mealy machines, Applications and limitations of FA.	6	CO2

		Self-learning Topics: State Elimination Method for converting FA to RE, Minimization of DFA using Equivalence Theorem, Conversion of Moore to Mealy & Mealy to Moore machine.		
III	Regular and Context Free Grammars	Grammars and Chomsky Hierarchy	1	CO3
		Regular Grammar (RG), Equivalence of Left and Right linear grammar, Equivalence of RG and FA.	2	CO3
		Context Free Grammars (CFG) Definition, Sentential forms, Leftmost and Rightmost derivations, Parse tree, Ambiguity, Simplification of CFG: Eliminating unit productions, useless production, useless symbols, and ϵ -productions, Normal Forms: Chomsky Normal Form (CNF) and Greibach Normal Form (GNF), Context Free language (CFL) - Application: Parser, Markup languages; Pumping lemma, Closure properties.	6	CO3
		Self-learning Topics: Left Recursion and Its Elimination, Applications of CFGs in XML Parsing, and Natural Language Processing (NLP).		
IV	Pushdown Automata (PDA)	Definition, Language of PDA, PDA as generator, decider and acceptor of CFG, Deterministic PDA , Non-Deterministic PDA, Equivalence of PDA and CFG, Application of PDA.	5	CO4
		Self-learning Topics: Parsing & PDA: Top-Down Parsing, Bottom-up Parsing, Closure properties and Deterministic PDA.		
V	Turing Machine (TM)	Definition, Design of TM as generator, decider and acceptor, Variants of TM: Multitrack, Multitape, Universal TM, Applications, Power and Limitations of TMs.	7	CO5
		Self-learning Topics: Algorithms using Turing Machine, The Model of Linear Bounded Automata		
VI	Decidability and Computability	Decidability and Undecidability, Recursive and Recursively Enumerable Language, Halting Problem, Rice’s Theorem, Post Correspondence Problem.	5	CO6
		Self-learning Topics: NP Completeness of the SAT Problem, A Restricted Satisfiability Problem		
Text Books:		1) John E. Hopcroft, Rajeev Motwani, Jeffery D. Ullman, Introduction to Automata Theory Language and Computation, 3rd Edition, Pearson Education, 2008. 2) Michael Sipser, Theory of Computation, 3rd Edition, Cengage learning. 2013. 3) Vivek Kulkarni, Theory of Computation, Illustrated Edition, Oxford University Press, (12 April 2013) India.		
References Books :		1) J. C. Martin, Introduction to Languages and the Theory of Computation, 4th Edition, Tata McGraw Hill Publication, 2013. 2) Kavi Mahesh, Theory of Computation: A Problem-Solving Approach, Kindle Edition, Wiley-India, 2011.		

Online References:	1) https://www.jflap.org/ 2) https://nptel.ac.in/courses/106104028 3) https://nptel.ac.in/courses/106104148
Internal Assessment (IA) :	Internal Assessment will consist of Two Compulsory IA Tests and shall be conducted for Total 40 Marks including 02 Tests of 20 marks each. Approximately 40% to 50% of syllabus content must be covered in First IA Test and remaining 40% to 50% of syllabus content must be covered in Second IA Test.
Question paper format:	• Question Paper will comprise of a total of six questions each carrying 20 marks Q.1 will be compulsory and should cover maximum contents of the syllabus • Remaining questions will be mixed in nature (part (a) and part (b) of each question must be from different modules. For example, if Q.2 has part (a) from Module 3 then part (b) must be from any other Module randomly selected from all the modules) • A total of Three questions needs to be answered